

Physical constants and conversion of units

Absolute zero temperature	-273.2°C
Acceleration due to gravity, g	9.807 m/s ²
Avogadro's number, N _A	6.022 x 10 ²³
Base of natural logarithms, e	2.718
Boltzmann's constant, k	1.381 x 10 ⁻²³ J/K
Faraday's constant k	9.648 x 10 ⁴ C/mol
Gas constant, $\bar{R}$	8.314 J/mol/K
Planck's constant, h	6.626 x 10 ⁻³⁴ J/s
Velocity of light in vacuum, c	2.998 x 10 ⁸ m/s
Volume of perfect gas at STP	22.41 x 10 ⁻³ m ³ /mol

Angle, θ	1 rad	57.30°
Density, ρ	1 lb/ft ³	16.03 kg/m ³
Diffusion Coefficient, D	1 cm ² /s	1.0 x 10 ⁻⁴ m ² /s
Energy, U	See opposite	
Force, F	1 kgf 1 lbf 1 dyne	9.807 N 4.448 N 1.0 x 10 ⁻⁵ N
Length, ℓ	1 ft 1 inch 1 Å	304.8 mm 25.40 mm 0.1 nm
Mass, M	1 tonne 1 short ton 1 long ton 1 lb mass	1000 kg 908 kg 1107 kg 0.454 kg
Power, P	See opposite	
Stress, σ	See opposite	
Specific Heat, Cp	1 cal/gal.°C Btu/lb.°F	4.188 kJ/kg.°C 4.187 kg/kg.°C
Stress Intensity, K _{Ic}	1 ksi $\sqrt{\text{in}}$	1.10 MN/m ^{3/2}
Surface Energy γ	1 erg/cm ²	1 mJ/m ²
Temperature, T	1°F	0.556°K
Thermal Conductivity λ	1 cal/s.cm.°C 1 Btu/h.ft.°F	418.8 W/m.°C 1.731 W/m.°C
Volume, V	1 Imperial gall 1 US gall	4.546 x 10 ⁻³ m ³ 3.785 x 10 ⁻³ m ³
Viscosity, η	1 poise 1 lb ft.s	0.1 N.s/m ² 0.1517 N.s/m ²

Conversion of units – stress and pressure*

	MPa	dyn/cm ²	lb.in ²	kgf/mm ²	bar	long ton/in ²
MPa	1	10 ⁷	1.45 x 10 ²	0.102	10	6.48 x 10 ⁻²
dyn/cm²	10 ⁻⁷	1	1.45 x 10 ⁻⁵	1.02 x 10 ⁻⁸	10 ⁻⁶	6.48 x 10 ⁻⁹
lb/in²	6.89 x 10 ⁻³	6.89 x 10 ⁴	1	703 x 10 ⁻⁴	6.89 x 10 ⁻²	4.46 x 10 ⁻⁴
kgf/mm²	9.81	9.81 x 10 ⁷	1.42 x 10 ³	1	98.1	63.5 x 10 ⁻²
bar	0.10	10 ⁶	14.48	1.02 x 10 ⁻²	1	6.48 x 10 ⁻³
long ton/ in²	15.44	1.54 x 10 ⁸	2.24 x 10 ³	1.54	1.54 x 10 ²	1

Conversion of units – energy*

	J	erg	cal	eV	Btu	ft lbf
J	1	10 ⁷	0.239	6.24 x 10 ¹⁸	9.48 x 10 ⁻⁴	0.738
erg	10 ⁻⁷	1	2.39 x 10 ⁻⁸	6.24 x 10 ¹¹	9.48 x 10 ⁻¹¹	7.38 x 10 ⁻⁸
cal	4.19	4.19 x 10 ⁷	1	2.61 x 10 ¹⁹	3.97 x 10 ⁻³	3.09
eV	1.60 x 10 ⁻¹⁹	1.60 x 10 ⁻¹²	3.38 x 10 ⁻²⁰	1	1.52 x 10 ⁻²²	1.18 x 10 ⁻¹⁹
Btu	1.06 x 10 ³	1.06 x 10 ¹⁰	2.52 x 10 ²	6.59 x 10 ²¹	1	7.78 x 10 ²
ft lbf	1.36	1.36 x 10 ⁷	0.324	8.46 x 10 ¹⁸	1.29 x 10 ⁻³	1

Conversion of units – power*

	kW (kJ/s)	erg/s	hp	ft lbf/s
kW (kJ/s)	1	10 ⁻¹⁰	1.34	7.38 x 10 ²
erg/s	10 ⁻¹⁰	1	1.34 x 10 ⁻¹⁰	7.38 x 10 ⁻⁸
hp	7.46 x 10 ⁻¹	7.46 x 10 ⁹	1	15.50 X 10 ²
Ft lbf/s	1.36 X 10 ⁻³	1.36 X 10 ⁷	1.82 X 10 ⁻³	1